

FITBUDDY AI

Phase 6: Project Testing

Prepared from the current public FitBuddy_ai repository and organized according to the Phase 1–6 structure of the provided AI/ML & GenAI project template.

6.1 Testing Objective

Testing verifies that the FitBuddy application behaves correctly from user input through database storage, dashboard display, feedback update and administrative inspection. The test plan below is designed around the routes and database fields visible in the current repository.

6.2 Functional Test Cases

Test ID	Scenario	Expected Result
TC-01	Open home page	Registration page loads successfully.
TC-02	Submit valid profile	User record and original plan are saved; dashboard opens.
TC-03	Open /user/{valid_id}	Correct stored profile is displayed.
TC-04	Open /user/{invalid_id}	HTTP 404 is returned with a clear error detail.
TC-05	Submit valid feedback	updated_plan and feedback_text are stored.
TC-06	Open /users	All stored users are displayed in the admin registry.
TC-07	Submit missing required form field	FastAPI validation prevents an invalid request from being processed normally.
TC-08	Restart application	Existing SQLite records remain available.

6.3 Database Tests

- Verify the users table is created when the application starts.
- Verify profile fields are inserted in the correct columns.
- Verify original_plan is non-empty after successful generation.
- Verify updated_plan and feedback_text are populated after feedback submission.
- Verify records remain after application restart.
- Verify admin ordering returns newest IDs first.

6.4 UI Tests

Page	Checks
Home	Labels, required fields, selectors, submit button and admin link are visible.
Dashboard	Profile metrics, nutrition card, plan, feedback form and navigation are visible.
Admin	Table headings, user metrics, original plan, updated plan and feedback are readable.

6.5 Integration Test

End-to-end scenario: register a profile → generate plan → confirm database record → open dashboard → submit feedback → confirm updated plan → open admin registry → verify both original and updated records. This is the primary demonstration path.

6.6 AI/GenAI Test Plan

Because the current public route constructs the workout plan directly, live Gemini behavior should be tested separately when the Gemini service is connected.

- Valid API key → model returns a valid structured plan.
- Invalid/missing API key → application shows a controlled configuration/API error.
- Model timeout or API failure → application uses controlled fallback handling.
- Unexpected model format → response validation rejects or safely normalizes the output.
- Feedback update → model receives the existing plan and feedback and returns a revised plan.

6.7 Defect Classification

Severity	Example
Critical	Application cannot start or database is corrupted.
High	Plan generation or feedback submission fails for valid input.
Medium	Admin page displays incomplete information.
Low	Minor styling, spacing or wording issue.

6.8 Acceptance Criteria

- Home page loads without server errors.
- Valid user data can be submitted and persisted.
- A plan is shown on the dashboard.
- Feedback can be stored and the updated plan is shown.
- Admin registry displays the stored records.
- Invalid user IDs are handled without an unhandled server failure.
- If Gemini is enabled, API failures and malformed model output are handled safely.
- Screenshots and test evidence are captured for the final submission.

6.9 Test Evidence Checklist

- Home page screenshot.
- Completed profile form screenshot.
- Generated dashboard screenshot.
- Feedback submission screenshot.
- Updated-plan screenshot.
- Admin registry screenshot.
- Database record screenshot or exported evidence.

- Gemini/API test evidence, if the GenAI service has been connected.

Project repository: https://github.com/asvikamoorthy-cyber/FitBuddy_ai

Reference structure: <https://github.com/Ravi-teja-777/AI-ML-and-GEN-AI-Track-Project-Template>

Scope note: This document describes the project state visible in the public repository and identifies proposed/future items separately where applicable.